Problem A. Counting 2

Time Limit 2000 ms

Problem Statement

There is a class with N students. The height of the i-th student $(1 \leq i \leq N)$ is A_i.

For each $j = 1, 2, \ldots, Q$, answer the following question.

- How many of the N students have a height of at least x_j?

Constraints

- $1 \leq N, Q \leq 2 \times 10^5$
- $1 \leq A_i \leq 10^9$
- $1 \leq x_j \leq 10^9$
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

$$
\begin{array}{l}
N \ \ Q \\
A_1 \ \ A_2 \ \ \ldots \ \ A_N \\
x_1 \\
x_2 \\
\vdots \\
x_Q
\end{array}
$$

Output

Print Q lines.

The j-th line $(1 \leq j \leq Q)$ should contain the number of students with a height of at least x_j.

Sample 1

Input	Output
3 1 100 160 130 120	2

The students with a height of at least 120 are the 2-nd and 3-rd ones.

Sample 2

Input	Output
5 5 1 2 3 4 5 6 5 4 3 2	0 1 2 3 4

Sample 3

Input	Output
5 5 804289384 846930887 681692778 714636916 957747794 424238336 719885387 649760493 596516650 189641422	5 3 5 5 5

Problem B. Distinct Numbers

Time Limit 1000 ms
Mem Limit 524288 kB

You are given a list of n integers, and your task is to calculate the number of *distinct* values in the list.

Input

The first input line has an integer n: the number of values.

The second line has n integers $x_1, x_2, \ldots, x_n$.

Output

Print one integers: the number of distinct values.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$

Example

Input	Output
5 2 3 2 2 3	2

Problem C. Sum of Two Values

Time Limit 1000 ms
Mem Limit 524288 kB

You are given an array of n integers, and your task is to find two values (at distinct positions) whose sum is x.

Input

The first input line has two integers n and x: the array size and the target sum.

The second line has n integers $a_1, a_2, \ldots, a_n$: the array values.

Output

Print two integers: the positions of the values. If there are several solutions, you may print any of them. If there are no solutions, print `IMPOSSIBLE`.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x, a_i \leq 10^9$

Example

Input	Output
4 8 2 7 5 1	2 4

Problem D. Apartments

Time Limit 1000 ms
Mem Limit 524288 kB

There are n applicants and m free apartments. Your task is to distribute the apartments so that as many applicants as possible will get an apartment.

Each applicant has a desired apartment size, and they will accept any apartment whose size is close enough to the desired size.

Input

The first input line has three integers n, m, and k: the number of applicants, the number of apartments, and the maximum allowed difference.

The next line contains n integers $a_1, a_2, \ldots, a_n$: the desired apartment size of each applicant. If the desired size of an applicant is x, they will accept any apartment whose size is between $x - k$ and $x + k$.

The last line contains m integers $b_1, b_2, \ldots, b_m$: the size of each apartment.

Output

Print one integer: the number of applicants who will get an apartment.

Constraints

- $1 \leq n, m \leq 2 \cdot 10^5$
- $0 \leq k \leq 10^9$
- $1 \leq a_i, b_i \leq 10^9$

Example

Input	Output
4 3 5 60 45 80 60 30 60 75	2

Problem E. Ferris Wheel

Time Limit 1000 ms
Mem Limit 524288 kB

There are n children who want to go to a Ferris wheel, and your task is to find a gondola for each child.

Each gondola may have one or two children in it, and in addition, the total weight in a gondola may not exceed x. You know the weight of every child.

What is the minimum number of gondolas needed for the children?

Input

The first input line contains two integers n and x: the number of children and the maximum allowed weight.

The next line contains n integers $p_1, p_2, \ldots, p_n$: the weight of each child.

Output

Print one integer: the minimum number of gondolas.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x \leq 10^9$
- $1 \leq p_i \leq x$

Example

Input	Output
4 10 7 2 3 9	3

Problem F. Sum of Three Values

Time Limit 1000 ms
Mem Limit 524288 kB

You are given an array of n integers, and your task is to find three values (at distinct positions) whose sum is x.

Input

The first input line has two integers n and x: the array size and the target sum.

The second line has n integers $a_1, a_2, \ldots, a_n$: the array values.

Output

Print three integers: the positions of the values. If there are several solutions, you may print any of them. If there are no solutions, print `IMPOSSIBLE`.

Constraints

- $1 \leq n \leq 5000$
- $1 \leq x, a_i \leq 10^9$

Example

Input	Output
4 8 2 7 5 1	1 3 4

Problem G. Playlist

Time Limit 1000 ms
Mem Limit 524288 kB

You are given a playlist of a radio station since its establishment. The playlist has a total of n songs.

What is the longest sequence of successive songs where each song is unique?

Input

The first input line contains an integer n: the number of songs.

The next line has n integers $k_1, k_2, \ldots, k_n$: the id number of each song.

Output

Print the length of the longest sequence of unique songs.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq k_i \leq 10^9$

Example

Input	Output
8 1 2 1 3 2 7 4 2	5

Problem H. Array Division

Time Limit 1000 ms
Mem Limit 524288 kB

You are given an array containing n positive integers.

Your task is to divide the array into k subarrays so that the maximum sum in a subarray is as small as possible.

Input

The first input line contains two integers n and k: the size of the array and the number of subarrays in the division.

The next line contains n integers $x_1, x_2, \ldots, x_n$: the contents of the array.

Output

Print one integer: the maximum sum in a subarray in the optimal division.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq k \leq n$
- $1 \leq x_i \leq 10^9$

Example

Input	Output
5 3 2 4 7 3 5	8

Explanation: An optimal division is $[2, 4], [7], [3, 5]$ where the sums of the subarrays are $6, 7, 8$. The largest sum is the last sum 8.

Problem I. Factory Machines

Time Limit 1000 ms
Mem Limit 524288 kB

A factory has n machines which can be used to make products. Your goal is to make a total of t products.

For each machine, you know the number of seconds it needs to make a single product. The machines can work simultaneously, and you can freely decide their schedule.

What is the shortest time needed to make t products?

Input

The first input line has two integers n and t: the number of machines and products.

The next line has n integers $k_1, k_2, \ldots, k_n$: the time needed to make a product using each machine.

Output

Print one integer: the minimum time needed to make t products.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq t \leq 10^9$
- $1 \leq k_i \leq 10^9$

Example

Input	Output
3 7 3 2 5	8

Explanation: Machine 1 makes two products, machine 2 makes four products and machine 3 makes one product.

Problem J. Subarray Sums I

Time Limit 1000 ms
Mem Limit 524288 kB

Given an array of n positive integers, your task is to count the number of subarrays having sum x.

Input

The first input line has two integers n and x: the size of the array and the target sum x.

The next line has n integers $a_1, a_2, \ldots, a_n$: the contents of the array.

Output

Print one integer: the required number of subarrays.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x, a_i \leq 10^9$

Example

Input	Output
5 7 2 4 1 2 7	3